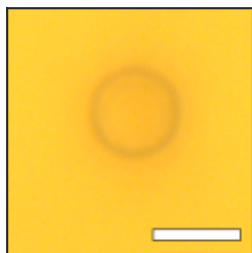


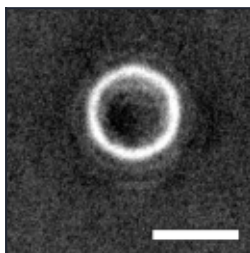
BF-Particle-Tracker

User guide and image-processing workflow | Generated 2026-05-12

A practical guide for loading microscopy videos, tuning visible preprocessing, tracking a particle centroid, and exporting trajectory data with metadata.



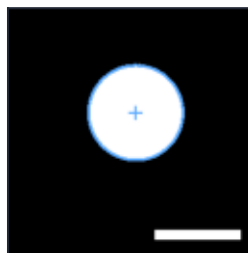
Raw image



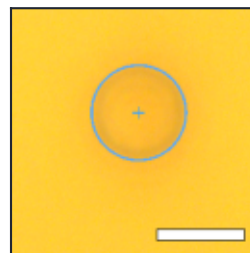
Preprocessed image



Binary image



Enhanced binary image



Tracking overlay

Core workflow

1. Load video stack

Open one TIFF/OME-TIFF file or combine split files into a single continuous stack.

2. Tune preprocessing

Use the four image panels to adjust contrast, background correction, and threshold behavior.

3. Segment binary blob

Clean the binary mask until the selected region represents the particle rather than noise.

4. Validate initial detection

Choose a reliable frame and set its centroid as the zero-reference for relative motion.

5. Track trajectory

Process the selected frame range while checking the overlay, speed, and rejection status.

6. Export results

Save frame-by-frame measurements in CSV and the current experiment settings in JSON.

Image-processing pipeline

From raw frame to cleaned binary region

Raw image

The selected frame is read from the TIFF stack. If a processing square is enabled, only that crop is passed to the image-processing and tracking pipeline.

Preprocessing

Intensity operations improve contrast between the particle and the background. Available operations include grayscale conversion, background correction, smoothing, top-hat filtering, inversion, percentile normalization, gamma correction, and CLAHE.

Binary segmentation

The preprocessed image is thresholded. Otsu mode estimates the cutoff automatically; manual mode lets the user choose the cutoff directly.

Morphological cleanup

Small particles, holes, and rough boundaries are cleaned using object-size filtering, opening, closing, and hole filling.

Blob measurement

Connected regions are measured with region properties. The selected blob provides the centroid, equivalent radius, area, eccentricity, and axis lengths.

Tracking

The previous particle position guides the next frame. The trajectory is stored in absolute and relative coordinates, with the initial detection as the reference point.

Acquisition and import controls

Metadata, calibration, loading, and processing region

Particle ID

A sample identifier saved with the results. It helps link trajectories to materials, particles, or experimental conditions.

Particle diameter

Physical diameter used to convert pixels into micrometers from the detected radius. Changing it rescales positions, radius, sigma, and stiffness outputs.

Frame interval

Time between frames. It defines the time axis in exported data and time-series plots.

Temperature

Used in the equipartition stiffness calculation through kBT. Higher temperature increases the reported stiffness for the same positional variance.

Frames to track

Controls how much of the stack is processed. Longer tracks improve statistics but take more time and are more sensitive to drift or loss of detection.

Wavenumber, power, note

Experimental metadata saved in the JSON file. These fields do not change the detection, but they make later comparison between experiments easier.

Fast lazy import

Opens large stacks quickly by reading frames only when needed. This makes browsing faster; tracking can still preload the selected frame range for speed.

Multiple-file import

Combines split files into a single virtual video when frame dimensions match. This is useful when microscope acquisitions are saved in parts.

Processing square

Restricts computation to a square crop. This can speed up analysis and reduce false detections from distant artifacts.

Square position and size

Moving the square changes the region analyzed. A smaller square is faster and cleaner, but must remain large enough to contain the particle throughout tracking.

Preprocessing controls

How each image-enhancement control affects segmentation

Convert to grayscale

Uses a single intensity channel for processing.
Keep enabled for most bright-field or grayscale microscopy data.

Background correction

Removes slow illumination variation.
Subtraction highlights local contrast; division compensates shading and uneven illumination.

Background sigma

Sets the spatial scale of the estimated background. A larger scale preserves broad particle features; a smaller scale removes more local structure.

Gaussian smoothing

Suppresses high-frequency noise before segmentation. Too much smoothing can soften the particle edge.

Top-hat

Enhances local objects relative to background.
The radius controls the size of features treated as background.

Invert image

Swaps dark and bright features. Use when the particle becomes easier to segment after contrast inversion.

Percentile normalization

Clips extreme intensities before rescaling.
This stabilizes contrast, but very strong clipping can saturate useful detail.

Gamma correction

Changes mid-tone contrast. Lower gamma brightens weak structures; higher gamma emphasizes strong bright regions.

CLAHE

Applies local adaptive contrast enhancement. It can reveal weak particles, but excessive local contrast may amplify noise.

Blob, binary, and tracking controls

Segmentation cleanup, temporal tracking, plots, and exports

Blob top-hat radius

Controls background removal in the blob-specific pipeline. It changes how strongly the particle is isolated before thresholding.

Invert before top-hat

Useful when the particle is dark in the raw image but should become bright for segmentation.

Threshold method

Otsu is automatic and convenient; manual mode is useful when background, halos, or noise confuse the automatic threshold.

Manual threshold

Foreground cutoff used in manual mode. Raising it keeps stronger pixels only; lowering it includes weaker signal and possibly noise.

Area limits

Reject objects that are too small or too large. This helps remove dust, fragments, merged blobs, or background artifacts.

Opening radius

Removes small protrusions and isolated pixels. Too much opening can erode the particle.

Closing radius

Fills small gaps and smooths boundaries. Too much closing can merge nearby regions.

Hole filling

Fills internal holes to create a more compact blob, often closer to a disk-like particle.

Choose by prediction

During tracking, selects the region nearest to the previous position instead of relying only on size.

Initial detection

Defines the reference point. Relative x and y trajectories are computed from this initial validated position.